Yaswanth Kalyan Ponugoti

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SUMMARY

Machine Learning Engineer and Applied Data Scientist with 4+ years of experience delivering production AI, predictive analytics, and data products across manufacturing, logistics, marketing, and healthcare. Builds end-to-end ML systems, RAG applications, recommendation engines, optimization models, and executive analytics using Python, SQL, BigQuery, Databricks, Power BI, and modern cloud tooling. Translates ambiguous operational problems into scalable solutions with measurable cost, yield, safety, and productivity outcomes while partnering directly with business, engineering, and frontline users.

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer | Tyson Foods Jun 2025 - Present · Springdale, AR

- Architected and delivered an AI-enabled CMMS platform using Next.js, React, TypeScript, Python, and BigQuery to replace a third-party system, creating a path to approximately $2M in avoided SaaS cost while unifying maintenance, labor, parts, and equipment workflows.
- Embedded retrieval-augmented generation (RAG), semantic search, and LLM-generated maintenance insights grounded in four operational sources: work orders, equipment history, technician activity, and materials; cut analysis turnaround time by approximately 50%.
- Independently owned the Ishida giveaway-optimization project from data ingestion and feature engineering through model evaluation, deployment, and monitoring; improved giveaway performance by 6%, saving 100K-150K pounds of product annually and approximately $200K per plant.
- Built a cross-plant recommendation system spanning three maintenance knowledge domains: preventive-maintenance history, technical documents, and work-order outcomes; surfaced evidence-backed tasks and strategies for similar equipment.
- Developed an AI-assisted manufacturer entity-resolution model using embeddings, fuzzy matching, and LLM validation to standardize vendor records and connect sourcing decisions to three reliability signals: downtime, repair cost, and maintenance history.
- Built Power BI dashboards for executive visibility across five operating areas: downtime, equipment performance, PM completion, labor capture, and maintenance cost, with plant, line, shift, and date drilldowns plus PDF exports.
- Engineered and validated production-grade SAP and manufacturing data products in BigQuery, improving downstream analytics and ML signal quality by approximately 30-40% through event sequencing, time-series alignment, and automated quality checks.
- Led delivery across five integrated workflows: maintenance planning, equipment reliability, labor analytics, parts visibility, and executive reporting; translated plant-user feedback into reusable components and prioritized product releases.

AI Engineer | Global Apps Mar 2025 - Jun 2025 · Santa Clara, CA

- Built reusable ML pipelines for two marketing use cases, campaign optimization and audience targeting, and developed LLM-powered content-generation and personalization capabilities.
- Used Databricks to prepare, validate, and version campaign datasets supporting repeatable model training, evaluation, and cloud-based inference workflows.
- Partnered with product and engineering teams to take AI features from prototype to user-facing release within a four-month delivery window.

Healthcare Research Assistant | Purdue University Aug 2024 - Dec 2024 · Indiana

- Developed Bayesian and machine learning models for two related research areas-diabetes and cardiometabolic disease-using large biomedical and genomic datasets.
- Built reproducible genomic processing and evaluation pipelines, reducing research turnaround time by approximately 40%.

Data Scientist / Senior Data Scientist | Schlumberger Jul 2021 - Jan 2023 · India

- Led a GCP and Looker Studio domestic logistics planning solution from proof of concept through MVP, translating operational constraints into decision-ready KPIs and contributing an estimated $800K in savings.
- Developed a global SAP material-price correction and data-quality solution that detected outliers across plants, countries, and materials, contributing nearly $1M in savings.
- Co-developed Redzone Notifier with rig-floor teams, applying video intelligence to detect safety risks and trigger real-time alerts as part of a two-product computer-vision portfolio.

- Built and deployed an Analog Gauge Reading pipeline to extract equipment values from video; together, the two computer-vision solutions supported a reported 30% improvement in HSE performance.
- Drove delivery across four analytics and AI products in three domains: logistics, procurement, and HSE; aligned data owners and domain experts from problem framing through MVP validation and adoption.

Machine Learning Engineer Intern | Shiash Info Solutions Jul 2020 - Jun 2021 · Chennai, India

- Developed crop disease prediction and fertilizer recommendation models and built an agricultural application with FastAPI.
- Designed two production-style ML services for crop-disease classification and fertilizer recommendations, adding reusable preprocessing, input validation, and REST inference endpoints for consistent application delivery.

Data Science Engineer Intern | Agrometrics Analytics & Services May 2019 - Dec 2019 · Hyderabad, India

- Developed U-Net and YOLO models for agricultural image analysis, achieving approximately 90% accuracy, and built data preprocessing pipelines.
- Prepared and validated image datasets through annotation checks, augmentation, and train-validation splits, enabling repeatable experiments across two agricultural computer-vision tasks.

TECHNICAL COMPETENCIES

Programming and application engineering Python, SQL, R, TypeScript, JavaScript, React, Next.js, FastAPI, Flask, REST APIs, Git, Linux, C++, Java

Machine learning and generative AI RAG, LLMs, agentic workflows, recommendation systems, optimization, entity resolution, NLP, computer vision, time-series modeling, Bayesian methods, scikit-learn, TensorFlow, PyTorch, Hugging Face

Data and cloud platforms BigQuery, Databricks, dbt, GCP, Azure, Spark, PostgreSQL, MongoDB, Cassandra, ETL/ELT, dimensional modeling, data quality, feature engineering

MLOps and analytics MLflow, model evaluation, experiment tracking, inference pipelines, monitoring, Docker, Kubernetes, Power BI, Tableau, Looker Studio, Plotly, Seaborn, Matplotlib, Pandas, NumPy

EDUCATION

M.S. Computational Data Science | Purdue University Dec 2024

Computer Science and Mathematics · Dean's Scholarship · Coursework: Bayesian Statistics, Categorical Data Analysis, Statistical Machine Learning, Statistical Quality Control

B.Tech. Computer Science, Specialization in Data Analytics | Vellore Institute of Technology Jun 2021

PERSONAL PROJECTS (SELECTED)

- **FMGE Mock Exam and AI Study Assistant** (GitHub): Built two AI workflows, question generation and RAG-based study assistance, using embeddings, semantic retrieval, a local LLM, React/Vite, and Supabase to deliver grounded explanations and timed practice tests.
- **YouTube Video Summarizing Agent:** Developed a two-stage Gemini, LangChain, and Groq LLaMA workflow that summarizes long-form videos and supports context-aware follow-up questions.
- **Metaphor Detection** (GitHub): Fine-tuned and evaluated a DistilBERT NLP classifier for metaphor recognition, achieving a reported validation loss of 0.045.
- **Crop Row and Leaf Segmentation:** Developed two U-Net computer-vision models for crop-row and individual-leaf segmentation, achieving 85% and 97% accuracy, respectively.
- **Bayesian Ad Conversion Testing:** Compared two advertising variants with Bayesian modeling, quantified uncertainty in conversion lift, and translated posterior results into campaign decisions.
- **Optimal Portfolio** (GitHub): Combined three probabilistic methods: Bayesian modeling, geometric Brownian motion, and MCMC; simulated stock behavior and evaluated portfolio risk.

CERTIFICATIONS

- Microsoft Azure AI Fundamentals: AI Overview; Career Essentials in Generative AI - Microsoft and LinkedIn
- Advanced Computer Vision with TensorFlow - Coursera; IBM Python for Data Science, AI & Development
- Deep Learning Certificate - GUVI; Atlassian Agile Project Management Professional Certificate
- Python for Data Science - DataCamp; MATLAB Onramp - MathWorks